RedKross Cybersecurity

Cybersecurity Subdomain — Architecture, Content, Projects and Developer Handover

Target domain	https://cyber.redkross.org.in (proposed)
Document purpose	Build the specialist technical website for cybersecurity research, projects, labs, Academy, resources and cyber opportunities.
Version	Developer handover specification · Draft 1.0
Date	24 October 2026

Prepared for the developer building the two new RedKross websites. The developer should audit the supplied source folder and verify all live content before migration or publication.

Important: This document defines the target experience and content. It does not authorise production server changes, DNS changes or publication of unverified claims.

1. Executive brief

Build a specialist cybersecurity website for RedKross Research Foundation. It should consolidate the existing main site's technical research, project portfolio, lab concepts, Academy roadmap, technical resources and cybersecurity-specific opportunities.

The Cybersecurity site must be technically credible and honest about project maturity. It should not be a generic marketing site or present planned labs, courses or prototypes as operational services.

Website	Role	Boundary
Main: redkross.org.in	Parent organisation, mission, both pillars, governance, partnerships and unified opportunity discovery.	Link back to foundation.
Cyber: cyber.redkross.org.in	Technical research, projects, labs, Academy, resources, responsible disclosure and cyber opportunities.	Build in this project; hostname is proposed.
Psychology: psychology.redkross.org.in	Psychology services, research, training and psychology-specific opportunities.	Existing site; link to it and do not disrupt it.

All current project names, repositories, demos, status labels, performance claims and lab availability must be checked against the supplied source and live implementation before publication.

2. Goals and audiences

- Present RedKross as a technical research and engineering pillar of the foundation.
- Make real cybersecurity and AI-related projects discoverable with meaningful technical detail.
- Separate research interests, prototypes, beta projects, deployed tools and validated outputs.
- Provide clear technical opportunities and college/university partnership information.
- Explain the current state of labs and Academy without implying unavailable access.
- Support responsible security research, authorised testing and safe vulnerability reporting.
- Deliver an independent application folder ready for later deployment.

Primary audiences: cybersecurity students, interns, researchers, developers, technical collaborators, college/university departments, open-source contributors and organisations seeking research collaboration.

3. Navigation and sitemap

Recommended navigation: Home · About · Research · Projects · Labs & Cyber Range · Academy · Opportunities · Resources · Responsible Disclosure. Add a visible link to the main RedKross Foundation website. Provide a contextual Psychology link where relevant and in the footer.

Route	Required purpose/content
/	Cybersecurity homepage: mission, research areas, featured projects, labs status, Academy, opportunities and disclosure.
/about	Cybersecurity pillar's purpose, scope and relationship to the foundation.
/research	Technical research overview and cards for the research areas.
/research/cybersecurity	Security engineering, authorised assessment and defensive work.
/research/ai-security	AI system security, robustness and security implications of AI-enabled systems.
/research/digital-forensics	Evidence integrity, forensic analysis and reporting methods.
/research/threat-intelligence	Security information, detection and investigation themes.
/research/iot-and-cloud	Connected devices, firmware, cloud and infrastructure security.

Route	Required purpose/content
/research/human-centred-security	Human behaviour, social engineering defence and security awareness.
/projects	Searchable/filterable project portfolio.
/projects/[slug]	Reusable detailed project template with status, implementation, methods, limitations and links.
/labs	Lab catalogue and availability status.
/labs/offensive-security	Authorised security assessment learning, only if content is ready.
/labs/blue-team	Monitoring, log analysis, threat hunting and defensive response themes.
/labs/digital-forensics	Forensic investigation and malware-analysis themes, subject to approved scope.
/academy	Academy overview and status.
/academy/learning-paths	Planned/available learning pathways with accurate details.
/academy/roadmap	Milestones and status; distinguish planned from operational.
/opportunities	Cybersecurity-specific roles, internships, project contributions and research opportunities.
/opportunities/[slug]	Reusable detailed opportunity page.
/college-partnerships	Technical detail for college/university cybersecurity training and project mentoring.
/resources	Technical guides, documentation, research notes and learning resources.
/publications	Verified technical publications, reports and papers.
/responsible-disclosure	Scope, reporting steps, safe boundaries and verified contact route.
/trust	Lab safety, privacy, data handling and research responsibility.
/contact	Technical research, project, partnership and security contact routing.

4. Cybersecurity homepage — section order

Order	Section	Required elements
1	Hero	Headline, concise technical mission, Explore Projects and Explore Research CTAs.
2	Research domains	Cybersecurity, AI security, digital forensics, threat intelligence, IoT/cloud and human-centred security.
3	Featured projects	Verified projects only, with status and repository/demo links if working.
4	Labs & Cyber Range	Planned vs available status; no false launch buttons.
5	Academy	Learning initiative and roadmap; clear enrolment status.
6	Technical resources	Real guides, documentation, articles or research notes.
7	Opportunities	Genuinely open cybersecurity internships, research and contribution routes.
8	College partnerships	Cybersecurity foundation training, project-level thinking and guided projects.
9	Responsible disclosure	Visible route to verified reporting policy/contact.
10	Foundation footer	Links to main site, Psychology, technical policies and contact.

5. Homepage — proposed copy for implementation

Hero Research. Build. Defend.

Exploring cybersecurity, artificial intelligence and digital forensics through technical research, practical engineering and responsible innovation.

RedKross Cybersecurity is the technical pillar of RedKross Research Foundation. We develop and document projects, investigate security challenges and work toward making technical knowledge more accessible.

Primary CTA: Explore Projects
Secondary CTA: Explore Research

Research Domains Security Across Systems, Data and People.

Cybersecurity & Security Engineering — Investigating digital security challenges through authorised testing, vulnerability analysis, secure engineering and defensive techniques.

AI & AI Security — Exploring the security and reliability of AI-enabled systems, including model robustness, adversarial risks and AI-assisted security workflows.

Digital Forensics & Incident Response — Developing knowledge and methods for analysing digital evidence, investigating incidents and documenting technical findings.

Threat Intelligence & Detection — Exploring how security information can be collected, assessed and used to support detection and investigation.

IoT, Cloud & Emerging Technologies — Investigating security challenges in connected devices, cloud infrastructure and evolving technology environments.

Human-Centred Security — Exploring human behaviour, social engineering, security decisions and defensive practices in collaboration with relevant disciplines.

Projects Engineering with Evidence.

Our public project portfolio includes tools and prototypes across AI, cybersecurity and related technologies. Each project should communicate its purpose, current maturity, technical approach and known limitations.

Browse project repositories, examine the implementation and explore opportunities to contribute where participation is available.

Labs Building a Responsible Cyber Research Environment.

We are developing a structured approach to practical security learning and experimentation. Planned lab environments are intended to support controlled, authorised exercises in offensive security, defensive operations and digital forensics.

Lab access will be announced when environments have been implemented, tested and approved for use.

Academy Learning Paths for Future Researchers.

The RedKross Academy is being developed as a structured learning initiative for cybersecurity and related technical disciplines. Curriculum planning, practical environments and assessment design will progress through staged development and validation.

Programme availability and enrolment information will be published when confirmed.

Opportunities Contribute to the Work.

Discover cybersecurity internships, research opportunities, engineering projects and open-source contributions when they are available.

Every opportunity should specify the skills required, expected commitment, supervision arrangements and application status.

Responsible Security Security Research Requires Responsibility.

We support authorised research, careful handling of sensitive information and clear communication of security findings. Our technical work should respect privacy, minimise harm and follow appropriate disclosure procedures.

Final CTA Explore the Research. Understand the Work.

Browse our technical projects, explore our research areas and follow the development of RedKross's cybersecurity research ecosystem.

6. About page — proposed copy

Page: /about

Technical Research for a Safer Digital World

RedKross Cybersecurity is the cybersecurity and technology pillar of RedKross Research Foundation. We focus on cybersecurity, artificial intelligence, digital forensics, security engineering and related technical research. Our work brings together investigation, experimentation, software development and technical documentation to explore practical security challenges.

What We Work On

Our technical interests span security automation, application and infrastructure security, AI-enabled security tools, digital evidence analysis, threat detection and emerging technologies. The scope of each project depends on its objectives, maturity and available resources.

How We Approach Research

We aim to define clear problems, investigate existing evidence, use appropriate methods, document implementation details and communicate limitations honestly. Security testing must be appropriately authorised and conducted within suitable boundaries.

Projects and Learning

Our public repositories provide a view of technical work across different stages of development. Some projects may be deployable, while others are prototypes or experiments. We are also developing structured technical learning and practical lab environments; availability will be announced as initiatives become operational.

Part of RedKross Research Foundation

RedKross Cybersecurity operates within the wider foundation, alongside Psychology & Human Wellbeing. We share a commitment to responsible research, accessibility, collaboration and evidence-informed work.

7. Research areas — content

Research landing page introduction: Our research interests span the systems, software, data and human factors that influence digital security. We aim to connect clear research questions with appropriate methods, reproducible experimentation and transparent technical reporting.

Research area	Content expected
Cybersecurity & security engineering	Vulnerabilities, security architecture, defensive controls, application security and authorised security automation.
AI security	Security and reliability of AI-enabled systems, model robustness, adversarial risks and safe AI application practices.
AI for security	Machine learning for anomaly detection, classification and security analysis; claims require appropriate evaluation.
Digital forensics	Evidence collection/preservation, digital analysis, integrity, provenance and reporting.
Threat intelligence & detection	Collection, assessment and use of security information to support detection and investigation.
IoT & cloud security	Connected devices, firmware, cloud configurations, identity and distributed infrastructure.

Research area	Content expected
Human-centred security	Human behaviour, online trust, social engineering defence and security awareness; link to Psychology where appropriate.

Each research card should show focus, current questions, related projects, verified publications (if any), related lab/learning path if available, and a clear status. Do not describe an area of interest as an active funded programme unless confirmed.

8. Project portfolio — required implementation

The current live portfolio includes examples such as DeepScan AI, FirmStrike, AppSec Orchestrator, RiskLens AI and Zero Trust Network Access work. These are migration candidates, not automatically approved publication claims. Verify repository details, status, working features, demo links and technical claims against the supplied code and live source before publishing.

Project listing introduction:

> Our project portfolio brings together technical experiments, research prototypes and software tools developed within the RedKross research and innovation ecosystem. Projects span cybersecurity, AI, digital forensics and related areas. Each listing should explain the problem it addresses, the approach taken, its current status and where its implementation can be examined.

Each project card must include: name; concise problem statement; research domain; maturity status; summary of working features; repository/demo links if verified; technology stack; last verified date.

Each project detail page must include:

- Problem statement, intended use and objectives.
- Current implementation and feature status.
- Architecture, methods and technologies.
- Testing approach and measured results only where documented.
- Known limitations, unresolved issues and dependencies.
- Safety, data handling, licensing and deployment considerations.
- Verified repository/documentation/demo links.
- Contributor attribution if confirmed, plus last verified date.

Example project	Starting content / verification instruction
DeepScan AI — Deepfake Detector	Existing site describes a deepfake-detection project and deployed components. Verify live demo, implementation, model description and any performance claims.
FirmStrike — IoT Firmware Security Analysis	Existing site describes an AI-assisted firmware analysis project. Verify whether scanning is live or simulated; label limitations prominently.
AppSec Orchestrator	Existing site describes code/document security scanning. Verify current engines, integrations and remediation claims.
RiskLens AI	Describe as a prototype unless its actual maturity has been verified; do not imply it replaces professional risk/compliance review.
Zero Trust Network Access work	Verify whether this is a project, concept, research note or deployed implementation before deciding its card status.

Do not invent GitHub URLs, statistics, detection rates, performance metrics, customer deployments or certifications. If evidence is missing, flag the item for owner approval.

9. Project maturity and status taxonomy

Status	Definition and rule
Planned	Concept or future work; not implemented. Do not show a working demo CTA.
In development	Implementation is underway; clearly identify incomplete parts.
Prototype	Proof of concept; not presented as production-ready.
Beta	Usable pre-release with known limitations and an appropriate warning.
Available / deployed	A verified accessible implementation exists and is maintained.
Evaluated / validated	Use only when documented evaluation supports the claim; explain method and scope.

The status should be visible on cards and detail pages. A public repository alone does not prove a project is operational or validated.

10. Labs & Cyber Range

Page: /labs

Proposed copy

Practical Security Research in Controlled Environments

RedKross is developing a structured approach to practical cybersecurity learning and experimentation. The proposed Cyber Range is intended to support controlled exercises that help learners and researchers explore security testing, defensive operations and digital forensics within authorised, appropriately isolated environments.

Planned Lab Areas:

- Offensive Security — planned learning environment for authorised assessment, vulnerability analysis and controlled attack simulation against deliberately vulnerable targets.
- Blue-Team Defence — planned environment for security monitoring, log analysis, threat hunting, incident investigation and defensive response.
- Digital Forensics & Malware Analysis — planned environment for digital evidence examination and suspicious-file analysis under appropriate controls.

Availability statement: These descriptions represent planned capabilities, not a claim that public lab environments are currently available. Access, supported exercises and participation requirements will be published when environments are implemented, tested and approved for use.

Use consistent Planned / In development / Available status labels. Do not provide nonfunctional Launch Lab buttons or claim public access without a working, approved environment.

11. Academy and learning pathways

Page: /academy

Proposed copy

The RedKross Academy is being developed as a structured learning initiative focused on cybersecurity, applied technology and related research disciplines. Our intended approach combines clear learning objectives, technical documentation, practical exercises and assessment appropriate to each programme.

Planned pathways may include Ethical Hacking & Cybersecurity; Digital Forensics & Incident Response; and Human-Centred Security. Each pathway must be labelled planned until curriculum, supervision, exercises and availability have been confirmed.

Programme availability: The Academy is in development. Planned pathways are not open for enrolment unless a programme is explicitly marked Available. Publish curriculum, eligibility, learning hours, assessment, fees and certification arrangements only after approval.

Do not promise certification, guaranteed employment or fixed course duration unless an approved programme can deliver it.

12. Cybersecurity opportunities

Page: /opportunities

Proposed copy

Contribute to Research and Engineering

RedKross Cybersecurity brings together technical projects, research interests and opportunities to develop practical skills. When positions or programmes are available, this page will provide clear information about the work, eligibility, supervision, expected commitment and application process.

- Cybersecurity internships: software engineering, security testing, AI-assisted tooling, digital forensics, documentation or research support depending on the approved project.
- Research participation: announce only when the project has an appropriate methodology, supervision and participation process.
- Open-source contributions: explain repository contribution rules, licence and expectations where contributions are actually accepted.
- College training and project programmes: provide technical details on the dedicated college partnerships page.

Only listings marked Open are accepting applications. A public repository does not automatically mean an internship exists. Psychology-specific opportunities should remain on the Psychology site.

13. College & university cybersecurity partnerships

Page: /college-partnerships

Proposed copy

From Classroom Knowledge to Practical Project Experience

RedKross Cybersecurity works toward bridging the gap between academic learning and the practical skills required to contribute to cybersecurity and technology projects. We invite colleges, universities and academic departments to explore partnerships that combine foundational cybersecurity training, project-level thinking, guided implementation and technical evaluation.

What students can learn

- Cybersecurity fundamentals: networking, operating systems, Linux, security principles, common vulnerabilities and defensive practices.
- Research and problem-solving: investigate a problem, review existing solutions, define requirements and justify technical decisions.
- Project development: planning, architecture, task allocation, version control, implementation, testing and documentation.
- Security tools and controlled labs: introductory experience subject to available infrastructure and authorised lab scope.
- Technical communication: demonstrate features, document results, explain design decisions and identify limitations.

Proposed collaboration formats

- Introductory cybersecurity workshops.

- Structured cybersecurity foundation training.
- Guided student project programmes and semester-linked technical projects.
- Faculty-supported mentoring and project reviews.
- Research and innovation collaboration.
- Advanced training pathways for suitable students when available.

Proposed student-development stages

1. Institutional consultation and needs assessment.
2. Foundational cybersecurity and safe-lab training.
3. Project planning, problem definition, requirements and milestones.
4. Guided implementation in authorised, controlled environments.
5. Evaluation, demonstration, documentation and feedback.

The format, curriculum, duration, fees, capacity, trainer availability, supervision and deliverables must be agreed with each institution. Do not promise job placements, internships, certifications or guaranteed outcomes by default.

CTA: Enquire About a College Tie-Up — link to a real institutional enquiry form on the main website or a functioning dedicated form.

14. Resources and publications

/resources — proposed copy

Explore technical guides, project documentation, research notes and educational resources published through RedKross's cybersecurity work. Resources should explain methods, tools and security concepts while making assumptions, scope and limitations clear.

- Technical guides — reviewed explanations of cybersecurity methods and defensive practices.
- Project documentation — installation, architecture, limitations and usage guidance.
- Research notes — technical observations and experimental documentation appropriate for public release.
- Learning resources — educational material for students and researchers.

/publications — proposed copy

This section may include research papers, technical reports, whitepapers, case studies and other reviewed materials when available. Each listing should identify authors, publication date, output type, review status, source and limitations.

Do not label a technical report, internal document, repository or preprint peer-reviewed unless the applicable review process has occurred. If there are no approved publications, show a concise status page rather than placeholder papers.

15. Responsible disclosure and trust

Page: /responsible-disclosure

Proposed copy

RedKross encourages responsible reporting of security vulnerabilities affecting systems within the scope of an applicable disclosure policy. Reports should provide enough information for assessment while avoiding unnecessary access to data, disruption or harm.

A report should include the affected asset, concise description, safe reproduction steps where appropriate, potential impact and relevant supporting evidence. Do not include passwords, private keys, personal data or other

unnecessary sensitive information.

Only test systems you own or have explicit authorisation to assess. Do not access, alter or disclose other people's data, disrupt services, deploy malware or exceed authorised scope.

Use the verified security reporting address or approved form. Before publication, the foundation must approve the in-scope assets, exclusions, safe-harbour language, reporting contact, acknowledgement process and remediation expectations.

16. Design, accessibility and technical requirements

Design direction

- Technical and research-oriented visual identity: dark navy, restrained blue accents, strong contrast and clean typography.
- Use genuine project screenshots, architecture diagrams and code examples only when available and approved; illustrative imagery must not be represented as actual product UI.
- Shared RedKross logo, typography, spacing, footer and accessibility conventions with the main site.
- Project cards should prioritise title, purpose, maturity status and verified links, not decorative metrics.

Accessibility and performance

- Responsive mobile/tablet/desktop layouts, semantic HTML, logical headings and keyboard navigation.
- Visible focus indicators, accessible forms, suitable alt text and reduced-motion support.
- Optimised assets; avoid excessive animation, trackers and third-party scripts.
- Clear loading, empty, success and error states.

Technical requirements

- Independently buildable project folder with source, package manifest, lockfile and README.
- Document framework and version after inspecting existing source; do not assume without audit.
- Provide .env.example with safe placeholders; no credentials or production secrets in code.
- Use structured project/opportunity content; no duplicate status or URLs across pages.
- Forms need real endpoints and server-side validation where applicable.
- SEO metadata, canonical URLs, sitemap/robots configuration and redirect map.
- No production DNS, Nginx, PM2, Cloudflare or server changes in this development phase.

17. Content migration map

Existing content	New destination	Action
Cybersecurity research areas	Cyber /research	Retain and expand technical descriptions; verify status.
Project/GitHub portfolio	Cyber /projects	Move detailed project listings and build detail pages.
Research labs	Cyber /labs	Consolidate labs; clearly label planned vs available.
Innovation Lab	Cyber /projects and Main /innovation	Split technical projects from foundation-wide innovation approach.
Academy roadmap	Cyber /academy/roadmap	Preserve useful milestones; verify current state.
Cyber internships	Cyber /opportunities	Verify open/closed status and application routes.
Technical guides/reports	Cyber /resources and /publications	Publish only approved, verified resources.
Responsible disclosure	Cyber /responsible-disclosure	Verify policy, scope and contact.
Foundation identity	Main website	Link back to main rather than duplicate all About content.
Psychology content	Existing Psychology site	Do not migrate; link contextually where appropriate.

18. Acceptance criteria and handover

- Cyber homepage and agreed pages are complete and populated with verified content.
- Project catalogue has real repository links, accurate maturity status and no fabricated claims.
- Research pages distinguish research interests from active programmes and validated outputs.
- Planned labs and Academy pathways are not presented as available.
- Cyber opportunity listings have correct status, eligibility, canonical destination and functioning application route.
- College partnership page explains training, project-level thinking, delivery stages and enquiry route.
- Responsible disclosure contact and policy scope are verified before publication.
- Main foundation and existing Psychology sites are linked correctly.
- Responsive layouts, accessibility, links, forms and production build are tested.
- Handover includes source folder, README, .env.example, dependencies, testing report, migration notes and unresolved-issue list.

Suggested folder structure:

redkross-website-handover/\n main-website/\n cybersecurity-website/\n documentation/page-inventory.md\n documentation/content-migration-map.md\n documentation/legacy-url-redirects.md\n documentation/testing-report.md\n documentation/deployment-notes.md\n HANDOVER.md

Recommended sequence: audit existing source; approve content ownership; design shared identity and distinct visual styles; build both sites; test; hand over. Deployment and migration will happen later as a separate phase.

19. Approval items before publication

- Confirm cyber.redkross.org.in as final hostname.
- Audit supplied source folder and verify every repository/demo and feature claim.
- Confirm project maturity labels, lab availability and Academy status.

- Confirm which cyber opportunities are genuinely open and who maintains them.
- Approve college programme commitments, trainer availability, fees and delivery model before making specific promises.
- Approve responsible-disclosure scope, reporting address and process.
- Confirm contact details, privacy wording and any policy or organisational claims.

This specification is a development baseline. The developer must verify live facts and supplied source before publication. Proposed copy is not evidence that a project, policy, lab or programme is operational.